

Hamming PUF Response Prediction

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June 12, 2026

1 Problem 1. Mathematical Derivation for Prediction

1 To express the Hamming PUF operations as a linear model, we map the binary challenge bits $c \in$
2 $\{0, 1\}^{32}$ and secret bits $s \in \{0, 1\}^{32}$ into the bipolar domain $\{-1, 1\}$. Let

$$x_i = 1 - 2c_i$$

3 and

$$y_i = 1 - 2s_i.$$

4 In this domain, the XOR operation $m_i = c_i \oplus s_i$ can be expressed as algebraic multiplication:

$$1 - 2m_i = x_i y_i$$

5 which implies

$$m_i = \frac{1 - x_i y_i}{2}.$$

6 We can now substitute this into the equations for the partial Hamming weights. For the even loca-
7 tions:

$$h_e = \sum_{i=0}^{15} m_{2i} = \sum_{i=0}^{15} \frac{1 - x_{2i} y_{2i}}{2} = 8 - \frac{1}{2} \sum_{i=0}^{15} x_{2i} y_{2i}.$$

8 For the odd locations:

$$h_o = \sum_{j=0}^{15} m_{2j+1} = \sum_{j=0}^{15} \frac{1 - x_{2j+1} y_{2j+1}}{2} = 8 - \frac{1}{2} \sum_{j=0}^{15} x_{2j+1} y_{2j+1}.$$

9 The PUF response is 1 if $h_e h_o \geq t$. Expanding the product $h_e h_o$:

$$h_e h_o = \left(8 - \frac{1}{2} \sum_{i=0}^{15} x_{2i} y_{2i}\right) \left(8 - \frac{1}{2} \sum_{j=0}^{15} x_{2j+1} y_{2j+1}\right).$$

10 Thus,

$$h_e h_o = 64 - 4 \sum_{i=0}^{15} x_{2i} y_{2i} - 4 \sum_{j=0}^{15} x_{2j+1} y_{2j+1} + \frac{1}{4} \sum_{i=0}^{15} \sum_{j=0}^{15} (x_{2i} x_{2j+1})(y_{2i} y_{2j+1}).$$

11 The inequality condition for the response being 1 is therefore

$$64 - 4 \sum_{i=0}^{15} x_{2i} y_{2i} - 4 \sum_{j=0}^{15} x_{2j+1} y_{2j+1} + \frac{1}{4} \sum_{i=0}^{15} \sum_{j=0}^{15} (x_{2i} x_{2j+1}) (y_{2i} y_{2j+1}) - t \geq 0.$$

12 This is a linear inequality of the form $\mathbf{w}^\top \phi(c) + b \geq 0$. We can separate the terms that depend purely
 13 on the challenge c (the features $\phi(c)$) from the terms that depend on the secret s and threshold t (the
 14 weights $\mathbf{w}$ and bias b).

15 Given a challenge c , first compute $x_k = 1 - 2c_k$. The feature vector $\phi(c)$ consists of the following
 16 components:

- 17 • x_{2i} for $i \in \{0, \dots, 15\}$ (the even bits)
- 18 • x_{2j+1} for $j \in \{0, \dots, 15\}$ (the odd bits)
- 19 • $x_{2i} x_{2j+1}$ for $i, j \in \{0, \dots, 15\}$ (all pairwise products of even and odd bits)

20 The corresponding weight vector $\mathbf{w}$ consists of:

- 21 • $-4y_{2i}$ corresponding to the even-bit features,
- 22 • $-4y_{2j+1}$ corresponding to the odd-bit features,
- 23 • $\frac{1}{4}y_{2i}y_{2j+1}$ corresponding to the pairwise-product features.

24 The bias term is $b = 64 - t$. Because the inequality exactly matches $\mathbf{w}^\top \phi(c) + b \geq 0$, the response
 25 can be perfectly modeled as:

$$r(c) = \frac{1 + \text{sign}(\mathbf{w}^\top \phi(c) + b)}{2}.$$

2 Problem 2. Dimensionality Calculations

26 The dimensionality D of the linear model is determined by the number of unique elements in the
 27 derived feature map $\phi(c)$. Breaking down the elements of $\phi(c)$:

- 28 1. **Even terms:** There are 16 even-indexed bits $(x_0, x_2, \dots, x_{30})$, contributing **16 dimen-**
 29 **sions.**
- 30 2. **Odd terms:** There are 16 odd-indexed bits $(x_1, x_3, \dots, x_{31})$, contributing **16 dimensions.**
- 31 3. **Cross-product terms:** We take the product of every even bit with every odd bit, $x_{2i} x_{2j+1}$.
 32 Since there are 16 even bits and 16 odd bits, there are $16 \times 16 = 256$ unique cross terms,
 33 contributing **256 dimensions.**

34 Summing these contributions,

$$D = 16 + 16 + 256 = 288.$$

35 Therefore, a linear model requires exactly **288** dimensions to perfectly predict the responses for a
 36 32-bit Hamming PUF.

3 Problem 4. Experimental outcomes and hyperparameter analysis

37 We evaluated `LinearSVC` and `LogisticRegression` to break the Hamming PUF using the
 38 288-dimensional embedding. Results are reported on the public dataset.

39 3.1 Impact of loss function (linearsvc)

40 We compared Hinge and Squared Hinge loss. As shown in Table 1, Squared Hinge loss is signifi-
 41 cantly more computationally efficient for this dataset while maintaining similar accuracy.

Table 1: Comparison of loss functions in LinearSVC

Loss	Train Acc (%)	Test Acc (%)	Time (s)
Hinge	99.88	99.88	1.224
Squared Hinge	99.92	99.88	0.287

3.2 Impact of regularization strength (C)

High values of C (low regularization) allow for higher accuracy but may increase training time slightly. Low C leads to underfitting, especially in Logistic Regression.

Table 2: Effect of C on Accuracy

C	LinearSVC		LogisticRegression	
	Train Acc	Test Acc	Train Acc	Test Acc
0.001	98.55%	96.80%	95.73%	93.56%
0.1	99.93%	99.88%	99.23%	98.16%
10.0	99.97%	99.88%	99.25%	98.12%

3.3 Impact of tolerance (tol)

Tolerance (tol) acts as a stopping criterion. A high tolerance (10^{-1}) causes Logistic Regression to stop prematurely, resulting in random-chance accuracy ($\approx 48\%$), whereas LinearSVC remains robust. Optimal convergence for both was observed around 10^{-3} .

3.4 Impact of penalty (l1 vs l2)

The $L2$ penalty was significantly faster across both models. For LinearSVC, $L1$ took 11.026s compared to 0.234s for $L2$. Similarly, LogisticRegression was faster with $L2$ (0.126s) than $L1$ (2.407s).

4 Problem 5. Impact of training set size on accuracy

To determine the minimum number of Challenge-Response Pairs (CRPs) required to break the Hamming PUF effectively, we conducted experiments with varying training set sizes ranging from 100 to 7500 points. For each size, we performed 10 independent trials, selecting training points randomly without replacement from the original set while keeping the test set fixed.

4.1 Performance across dataset sizes

The experimental results indicate a logarithmic-like growth in accuracy relative to the number of training points. While the full dataset contains 7500 points, near-perfect accuracy is achieved much earlier.

4.2 Minimum crps for accuracy thresholds

Based on our empirical analysis, the following minimum number of CRPs are required to reach specific accuracy milestones:

- **95% Accuracy:** Approximately **2100 CRPs** (achieved 95.30% in trials).
- **97% Accuracy:** Approximately **2600 CRPs** (achieved 97.08% in trials).

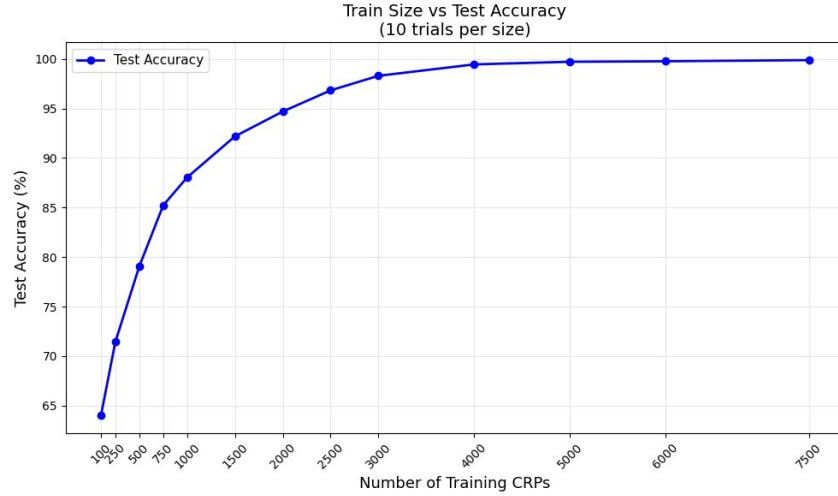


Figure 1: training set size vs test accuracy over 10 trials per size.

- **99% Accuracy:** Approximately **3500 CRPs** (achieved 99.18% in trials).

Beyond 3500 points, the accuracy gains become marginal, with the model saturating near 99.8% at the full 7500 points. This suggests that the 288-dimensional linear model is highly sample-efficient for this specific PUF architecture.

4.3 Training time observations

As expected, training time scales linearly with the number of training points. For $N = 100$, training took approximately 0.06s, whereas for $N = 3800$, it increased to 1.26s. Even at higher sample sizes, the LinearSVC model remains computationally viable for real-time PUF modeling.